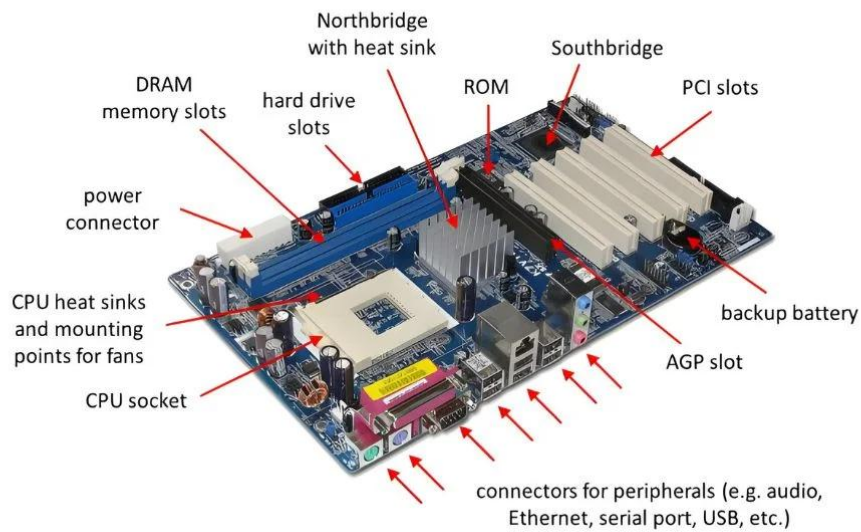


1.

Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).

Motherboard



2.

Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

1. AMD Socket AM5

- **Socket Type:** Land Grid Array (LGA-1718)
- **Description:** AMD's desktop platform designed for modern Ryzen processors, supporting DDR5 RAM and PCIe 5.0.
- **Compatible CPU:** **AMD Ryzen 7 7800X3D** (or Ryzen 9 9950X / Ryzen 5 7600)

2. Intel LGA 1700

- **Socket Type:** Land Grid Array (LGA-1700)
- **Description:** Intel's platform designed to support 12th (Alder Lake), 13th (Raptor Lake), and 14th Gen Core desktop processors with hybrid core architectures.
- **Compatible CPU:** **Intel Core i7-13700K** (or Core i5-12400 / Core i9-14900K)

3. AMD Socket AM4

- **Socket Type:** Pin Grid Array (PGA-1331)
- **Description:** AMD's legacy desktop socket that supported multiple generations of Zen architecture chips (Ryzen 1000 through 5000 series).
- **Compatible CPU:** **AMD Ryzen 7 5700X3D** (or Ryzen 5 5600X / Ryzen 9 5900X)

3.

Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

Think of the **CPU as the Boss** and the **Chipset as the Assistant**.

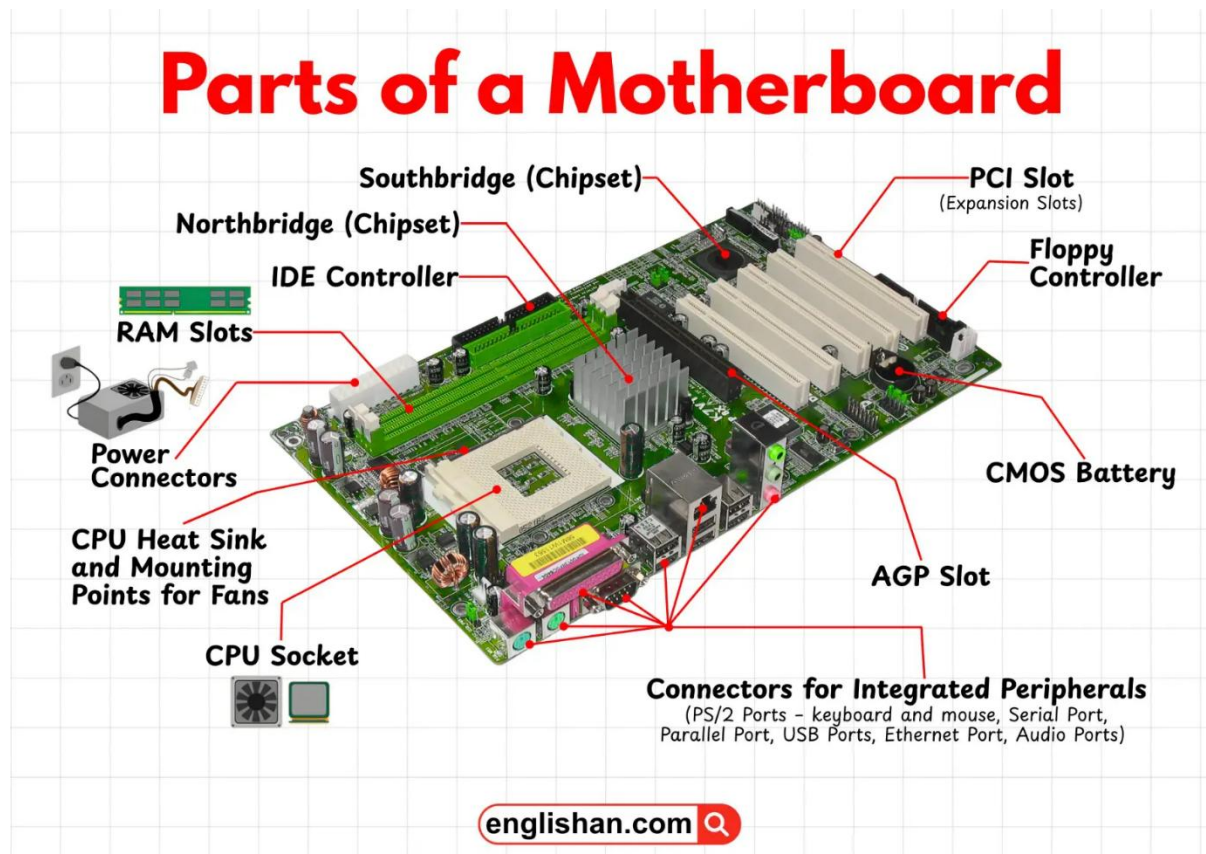
- **CPU (The Boss):** Does all the heavy thinking, makes decisions, and runs the business.
- **Chipset (The Assistant):** Handles communications. It brings mail from the door (mouse/keyboard), fetches files from the storage room (SSD), and delivers messages to other departments (screen/speakers).

What Happens When You Open Spotify

1. **You Click Spotify:** The mouse tells the **Assistant** (Chipset): *"The user clicked Spotify."*
2. **Alerting the Boss:** The **Assistant** runs to the **Boss** (CPU) and says: *"We need to open Spotify."*
3. **Getting the Files:** The **Boss** says: *"Go get the Spotify files from the storage room (SSD) and put them on my desk (RAM)."*
4. **Running the App:** The **Boss** reads the files, opens the app, and gets your playlists ready.
5. **Playing Music:** You press Play. The **Boss** translates the music file into sound and tells the **Assistant**: *"Send these sound signals to the speakers!"*

4.

Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.



- ☐ **CPU Socket:** Right in the middle-left area where the processor installs.
- ☐ **RAM Slots:** Long green slots next to the CPU socket for short-term memory sticks.
- ☐ **PCI / Expansion Slots:** Long white slots at the top-right used for graphic cards or sound cards.
- ☐ **Power Connectors:** White multi-pin sockets where cables from the power supply plug in.
- ☐ **CMOS Battery:** Small round silver battery that keeps the motherboard's system clock and BIOS settings saved even when turned off.

